MALE
FERTILITY

AMS®
America Medic & Science

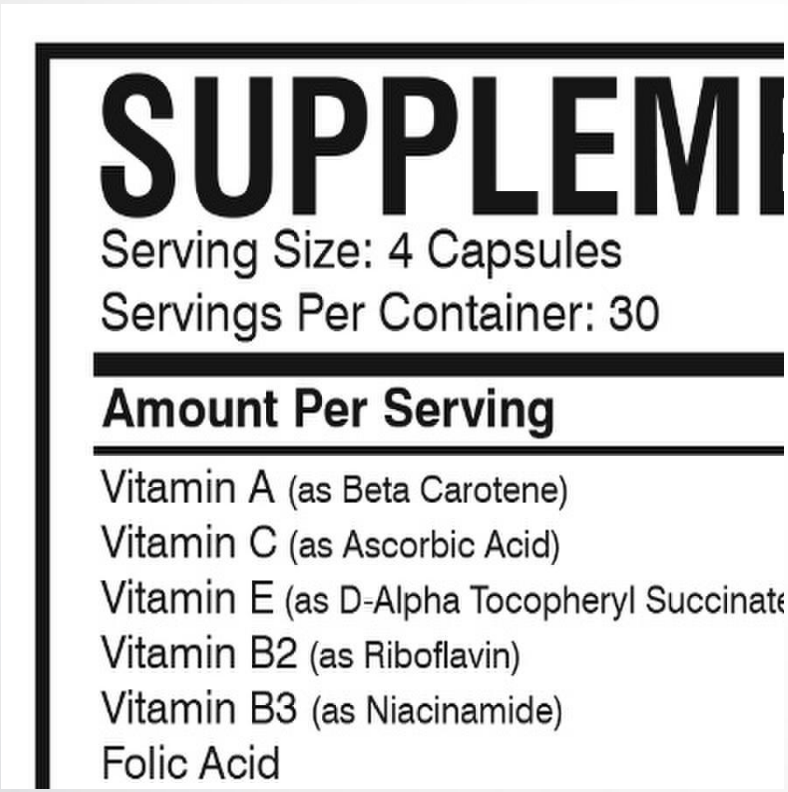
SUPREME UNIVERSAL FERTILITY

MFS Plus®

PHYSICIAN FORMULATED

Improved Formula

- ✓ Acetyl-L-Carnitine
- ✓ Co-Enzyme Q10
- ✓ Glutathione
- ✓ L-Methionine
- ✓ Lycopene
- ✓ Relora®



INDICATIONS

- Patients with normal hormonal assay with abnormal seminal fluid analysis
- Triple Defect Cases, OAT
- Idiopathic infertility (associated with normal semen analysis), High DFI.

BENEFITS

✓Enhances sperm motility
✓Increases sperm count
✓Enhances sperm morphology
✓Helps decrease elevated DNA Fragmentation Index
✓Super Antioxidant

Relora

Antioxidants + Vitamins

The Key ingredient in MFS formula is RELORA®

- Relora is a natural proprietary blend of patented (U.S. patent No. US 6,582,735)
- Relora contains the botanicals Phellodendrom amurense and Magnolia Officinalis
- Relora is approved by Dietary Supplement Health & Education regulated by FDA

The Key ingredient in MFS formula is RELORA®

- The supplement is widely known for its tremendous potential to help users lose weight and manage stress.
- Magnolia officinalis was widely used in the traditional Chinese Medicine for its neuroprotective and relaxing properties.
- Phellodendron amurense also originates in China where it was used for osteoarthritis, weight loss, diarrhea, stomach ulcers, and so on.

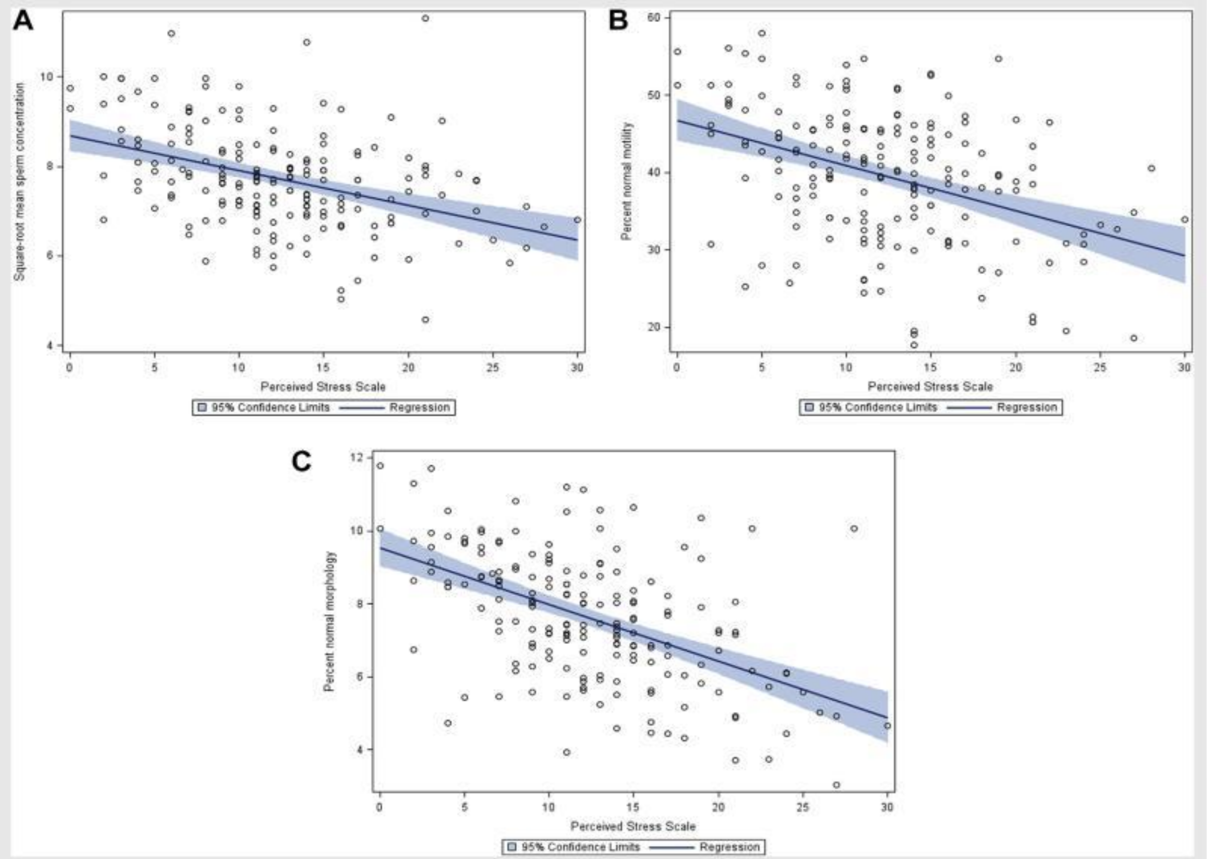
Why Relora in a fertility supplement?

- "There is now emerging evidence that male obesity impacts negatively on male reproductive potential not only reducing sperm quality, but in particular altering the physical and molecular structure of germ cells in the testes and ultimately mature sperm"

Spermatogenesis 2:4, 253-263;October/November/December2012; © 2012 Landes Bioscience

- The LIFE study : "We observed that adiposity is related to sperm production when assessed by BMI as well as WC. Overweight and obese men had a higher incidence of low ejaculate volume, semen concentration and total sperm count"

HumanReproduction,Vol.0,No.0pp.1–8,2013



Predicted mean for association between perceived stress score and sperm concentration (panel A), percentage of motile sperm (panel B), and percentage of morphologically normal sperm (panel C).

Janevic. Stress and semen quality. Fertil Steril 2014.

Janevic, T.,et al (2014). ***Effects of work and life stress on semen quality.*** *Fertility and Sterility, 102(2), 530-538. doi:10.1016/j.fertnstert.2014.04.021*

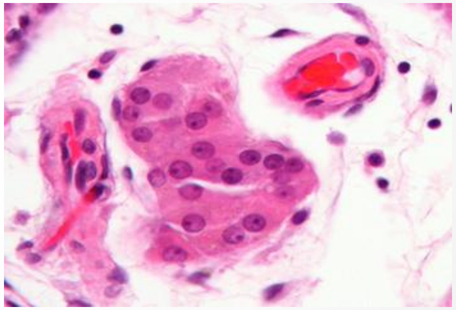
- Stress

Increase in serum glucocorticoid concentrations

Inhibition of testosterone-biosynthetic enzyme activity

Decreased rates of testosterone secretion

High Corticosterone* concentrations



The Leydig cell

Increase in their frequency of apoptosis

Hardy, M. P., et al(2005). Stress hormone and male reproductive function

MOA of Relora

- Relora alleviates stress and stress-related eating by decreasing cortisol
- Relora binds in CNS with chemical receptors other than BZD

↓

No sedation

- Relora decreases cortisol levels & peripheral sympathetic tone by activation noradrenal catabolism & its release by preventing rise by intracellular calcium ions

↓

Displacing the potentially harmful stress hormones

Net results, feel of alter relaxation

A trial was undertaken to measure cortisol and DHEA levels in patients with mild to moderate stress. Elevated cortisol levels and depressed DHEA levels are associated with chronic stress.

A two-week regimen of Relora® caused a significant (P= 0.003) increase in salivary DHEA (227%) and a significant (P = 0.01) decrease in morning salivary cortisol levels (37%). Cortisol and DHEA levels were returned to normal in all subjects during the course of Relora

- Open human trails (200mg*3) showed that Relora helps in controling stress related symptoms as irritability, emotional ups & downs, restlessness, tense muscle, poor sleep, and decrease appetite

- Stress management is important because stress exerts profound suppression on male reproductive system & also stress reduces sperm count & fecundity

- Sperm samples provided after more stimulations & less stress yield dramatically high motile sperm numbers

Alternative Therapies in Health and Medicine.
Effect of a proprietary Magnolia and Phellodendron extract on weight management: a pilot, double-blind, placebo-controlled clinical trial. 2006 Jan-Feb;12(1):50-4.

✓The purpose of this study was to determine the effects of Relora in overweight premenopausal women who reported higher than average anxiety and a tendency to eat more in response to stress.

✓ 250 mg, 3 times daily for 6 weeks

Subjects completed the study and Relora was well tolerated.

✓Significant weight gain during the study for the placebo group but no significant weight gain and some weight loss for the Relora group.

✓There was also a significant reduction in anxiety scores.

✓The Relora group presented a trend toward lower cortisol levels at the end of the study, particularly in the evenings. In contrast, evening cortisol levels increased in the control group.

The mechanism of action appears to be through reduction or normalization of stress hormone levels, and possibly perceived stress, thereby helping subjects maintain body weight

Effect of *Magnolia officinalis* and *Phellodendron amurense* (Relora®) on cortisol and psychological mood state in moderately stressed subjects

Talbott et al. Journal of the International Society of Sports Nutrition 2013, 10:37

- Randomized placebo controlled double-blind study design
- 250mg of Relora bid or a similar placebo regimen for 4 weeks
- Objective was to: explore how 4 weeks of relora supplementation (versus a placebo) affected cortisol, various moods and overall stress levels under conditions of moderate psychological stress.

No adverse events or side effects were reported

The MOA appears to be through reduction or normalization of stress

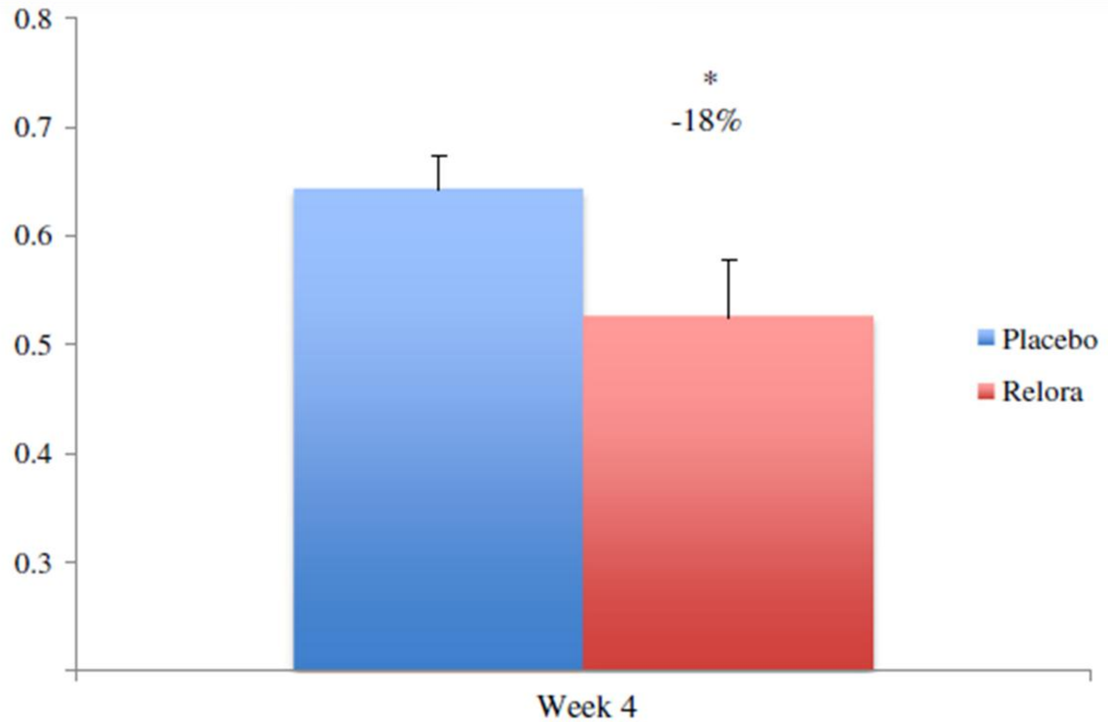


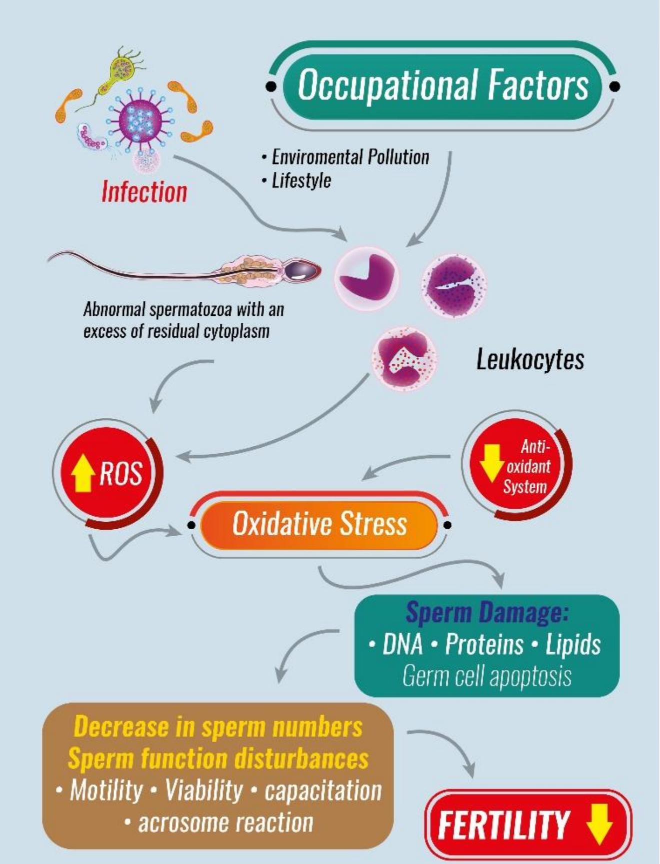
Figure 1 Salivary Cortisol (ug/ml). Salivary cortisol was 18% lower ($p<0.05$) in the Relora group compared to Placebo at Week 4 (0.525 ± 0.190 to 0.642 ± 0.353).

Talbott et al. Journal of the International Society of Sports Nutrition 2013, 10:37

- Relora is gentle on stomach & safe according to acute toxicity study(5g/kg) for 14 days that revealed no unwanted effects except mild diarrhea

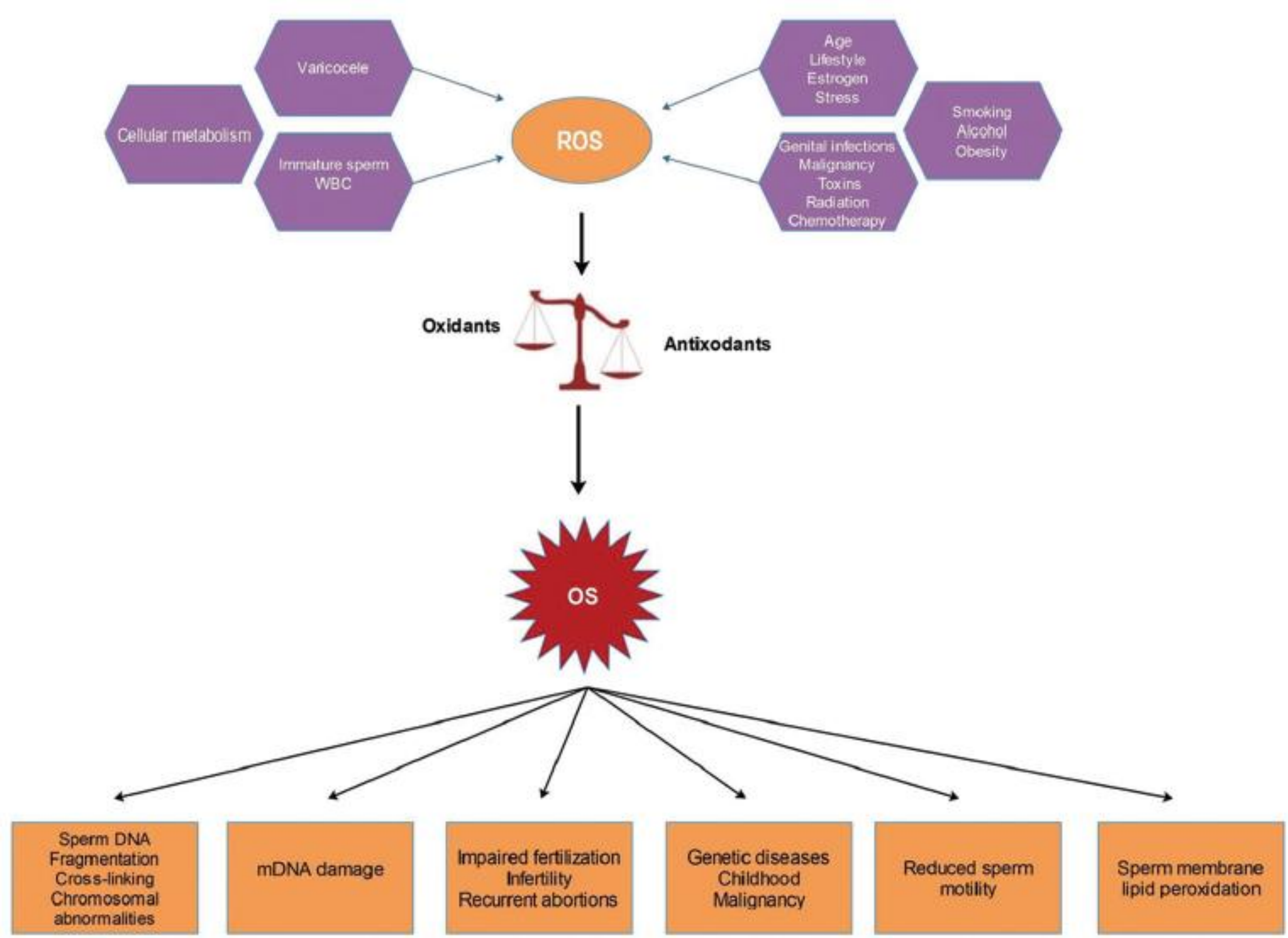
•Relora in US is used for weight loss that is associated with
stress & stress related condition known as metabolic
Syndrome
•Relora controls adrenal gland leading to a decrease in cortisol levels
& increase in DHEA → increase testosterone

- Spermatozoa are particularly vulnerable to the harmful effects of ROS, Oxidative stress:
  - **Affects their activity**
  - **Damages DNA structure**
  - **Accelerates apoptosis**



Walczak-Jedrzejowska, R., Wolski, J. K., & Slowikowska-Hilczer, J. (2013).Central European Journal of Urology, 66(1), 60.

Walczak-Jedrzejowska, R. et al, (2013). The role of oxidative stress and antioxidants in male fertility. Central European Journal of Urology, 66(1), 60.



Alahmar, A. T. et al (2019*). Journal of Human Reproductive Sciences, 12*(1), 4-18.

Crucial role of Antioxidants

- The protective antioxidant system in the semen is composed of enzymes, as well as non enzymatic substances, which closely interact with each other to ensure optimal protection against ROS

- **Non–enzymatic antioxidants**

Vitamins A, E, C, and B complex, glutathione, pantothenic acid, coenzyme Q10 and carnitine, and micronutrients such as zinc, selenium, and copper

- It seems that a deficiency of any of them can cause a decrease in total antioxidant status

- *In vitro* and *in vivo* studies demonstrate that many antioxidants possess a beneficial effect on fertility

Walczak-Jedrzejowska, R. et al, (2013).

"Effects of oral ´ antioxidant treatment upon the dynamics of human sperm DNA fragmentation and subpopulations of sperm with highly degraded DNA,"
C. Abad, M. J. et al., Andrologia, vol. 45, no. 3, pp. 211–216, 2013.

- Primary aim was to : **Determine the effect of oral antioxidant treatment (L-Carnitine; vitamin C; coenzyme Q10; vitamin E; zinc; vitamin B9; selenium; vitamin B12) for 3 months upon the dynamics of sperm DNA fragmentation in a cohort of 20 infertile patients diagnosed with Asthenoteratozoospermia**

- Secondary objective was to :Use the sperm chromatin dispersion test (SCD) to study antioxidant effects upon a specific subpopulation of highly DNA degraded sperm (DDS). Semen parameters and pregnancy rate (PR) were also determined

Results

- ✓ Significant improvement of DNA integrity at all incubation points
- ✓ The proportion of DDS* was also significantly reduced
- ✓ Semen analysis data showed a **significant increase in concentration, motility, vitality and morphology parameters**

- ✓ **Antioxidant treatment improves sperm quality not only in terms of key seminal parameters and basal DNA damage, but also helps to maintain DNA integrity**
- ✓ **Prior administration of antioxidants could therefore promote better outcomes following assisted reproductive techniques**

**DNA degraded sperm*

Review of Clinical Trials on Effects of Oral Antioxidants on Basic Semen and Other Parameters in Idiopathic Oligoasthenoteratozoospermia

Hindawi Publishing Corporation BioMed Research International Volume 2014, Article ID 426951

- Including reviews, controlled and randomized controlled clinical studies, 32 primary studies on idiopathic (OAT)
- Period between January 2000 and December 2013

✓ The majority of studies confirmed beneficial effect of different antioxidants on at least one of the semen parameters and the biggest effect was determined on sperm motility

✓ In many of these trials combinations of more than one antioxidant were assessed

✓ Most commonly antioxidants studied were vitamin E, vitamin C, selenium, CoQ10, N-acetyl-cysteine, L-carnitine, and zinc and their favorable effect was confirmed

Anti-Oxidant	Effects on iOAT	Oligospermia	Asthenospermia	Terato-spermia	DNA fragmentation
CoQ10	√	√	√	√	
Vit E	√	√	√		√*
Sel	√		√	√	√
Vit C	√				√*
N-A-C	√				
Zinc					√

*Combination of vitamin C and E showed the biggest favorable effect on DNA fragmentation

In conclusion, antioxidants play an important role in protecting semen from ROS and can improve basic sperm parameters in case of iOAT

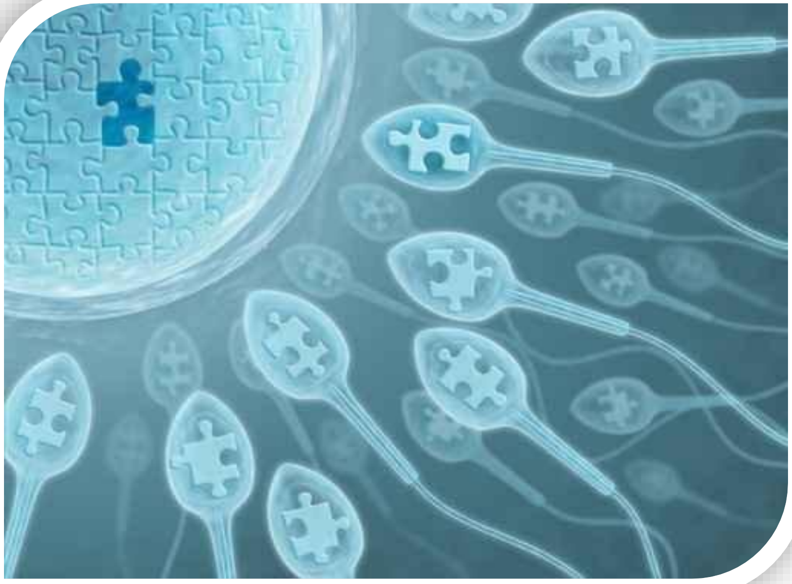
Ahmadi, S., et al (2016). Antioxidant supplements and semen parameters: An evidence based review. *Iranian Journal of Reproductive Medicine, 14*(12), 729.

- Approximately, 30-80% of infertility cases are caused by oxidative stress and decreased level of seminal total antioxidant capacity.

Anti-Oxidant	Sperm Count	Sperm Motility	Sperm Morphology	DNA damage
CoQ10	√			√
Vit E	√			√
Carnitine		√	√	

Effects of antioxidant treatment on DNA fragmentation Index

Medicine, Los Angeles, CA. (2004, October) Presented at the annual American Society for Reproductive Medicine in Philadelphia, PA.



✓Treatment of men with DFI > 30% with a 30 to 90 day course of antioxidant was associated with a statistically significant decrease in the DFI

Selenium–vitamin E supplementation in infertile men: effects on semen parameters and pregnancy rate

- The study consisted of a 14-week treatment phase.
- All patients were required to have ceased all medical therapy 6 weeks before study initiation.
- Eligible patients were assigned to fixed-dose treatment with 200 µg oral Se tablet (l-selenomethionine) daily in combination with 400 IU synthetic vitamin E (α-tocopherol).

Moslemi, M. K., et al (2011). *International Journal of General Medicine, 4*, 99.

Selenium–vitamin E supplementation in infertile men: effects on semen parameters and pregnancy rate

Table 2 Results of treatment on the basis of semen analysis improvement or drug-induced s

Quality of improvement	Motility improvement of at least 5%	Motility improvement of at least 10%	Improvement in motility and mor
Number of cases	144 cases (20.5%)	155 cases (22.5%)	42 cases (6%)
Semen analysis	Normal motility mean	Normal motility mean	Normal morpholog

Moslemi, M. K., et al (2011). *International Journal of General Medicine, 4*, 99.

Glutathione

- ✓ Master antioxidant, has synergistic effect with selenium
- ✓ Glutathione reduces the oxidative damage by neutralizing the harmful free radicals
- ✓ GSH concentration in the maturating spermatozoa gradually decline during spermatogenesis
- ✓ Thio-antioxidant system (mainly GSH) is 10 folds in seminal fluid more than plasma that show its effect & importance in spermatogenesis & decrease in DNA damage

Impact of Seminal Trace Element and Glutathione Levels on Semen Quality of Tunisian Infertile Men

Fatma Atig, et al Disclosures BMC Urol. 2012;12(6)

Study revealed that : Associations of seminal Zn, Se and GSH with other parameters of semen quality indicate that

- **The decrease of seminal antioxidants can be a risk factor for sperm abnormality, low sperm quality and idiopathic male infertility**
- **These data suggested that routine determination of antioxidant status during infertility investigation is recommended**

Acetyl-L-Carnitine

- ✓ L-carnitine has a pivotal role in cellular energy production, It is necessary for mitochondrial oxidation of long chain fatty acids. Carnitine also protects the cell membrane and DNA against damage induced by free oxygen radicals. It prevents protein oxidation and pyruvate and lactate oxidative damage

- ✓ The highest levels of L-carnitine in the human body are found in epididymal fluid, in which its concentration is around 2000 times higher than in circulating blood

Jarow, J. P. et al (2004). The Journal of Urology, 172(5), 2091-2092. doi:10.1016/S0022-5347(05)60950-0

Acetyl-L-Carnitine

- ✓ “The initiation of sperm motility occurs in parallel to an increase in carnitine concentration in the epididymal lumen and L-acetyl-carnitine in spermatozoa. Epididymal epithelium secretes L-carnitine into the lumen by specific active transport systems”

- ✓ “Taking into account their mechanisms and functions, Lcarnitine and L-acetyl-carnitine have been proposed and used as a possible treatment in selected forms of oligoasthenoteratozoospermia”

Jarow, J. P. et al (2004). The Journal of Urology, 172(5), 2091-2092. doi:10.1016/S0022-5347(05)60950-0

Acetyl-L-Carnitine

"A 2004 double-blind placebo controlled study in infertile men demonstrated that l-carnitine and acetyl l-carnitine significantly increased sperm concentration, motility and morphology in men with poor sperm quality, resulting in a significantly higher pregnancy rate"
(Cavallini, Ferraretti, et al.)

Acetyl-L-Carnitine

- From the publication Milestones in Medicine (Adis) : L-Carnitine & acetylcarnitine in male infertility:

- “These positive effects of both l-carnitine and acetyl-l carnitine on sperm quality were observed after at least 2 months of use, consistent with the 74-day sperm maturation cycle”

- “Improvements in sperm quality should be expected within 2 months of initiation of therapy, with maximum improvements after 6 months of therapy…”

Light and electron microscopic study of the effect of L-carnitine on the sperm morphology among sub fertile men

Middle East Fertility Society Journal (2010) 15, 95–105

- **Objective:** Was to evaluate the effect of L-carnitine on sperm morphology in sub fertile patients who need enhance for intra cytoplasmic sperm injection (ICSI) as a method of infertility treatment

According to the routine semen analysis, 85 patients were divided into:
Group 1: Ten normal fertile men
Group 2: 25 oligozoospermic cases with sperm count less than 20 million/ml
Group 3: 25 athenozoospermic cases with reduced sperm motility <40%
Group 4: 25 teratozoospermic cases with more than >40% abnormal forms

- L. carnitine therapy 1 gm /tab. b.i.d. for three months were given to groups 2, 3 and 4

Healthy group

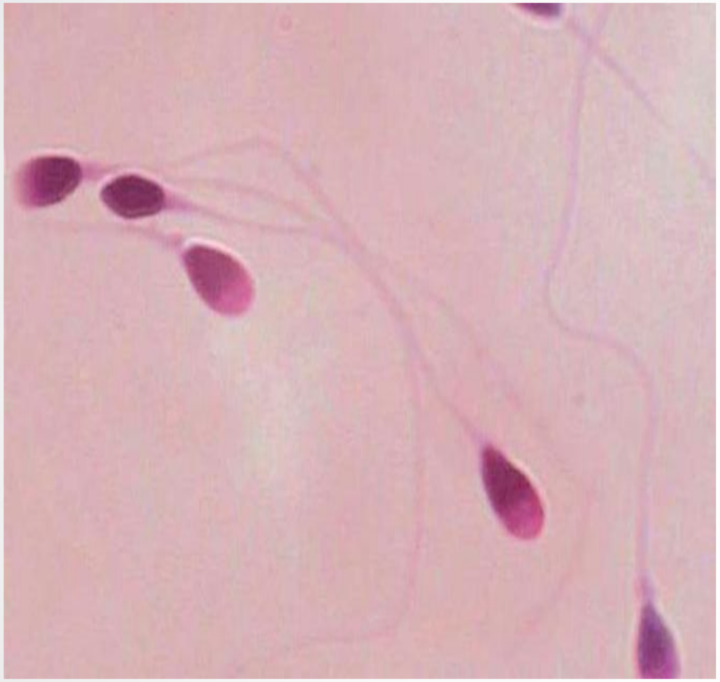


Figure 1 Photomicrograph of normal sperms from healthy fertile man.

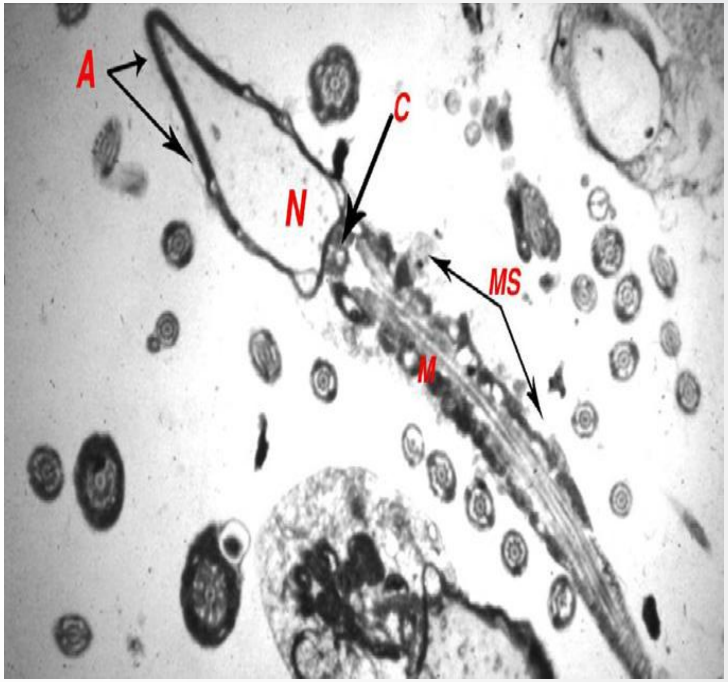
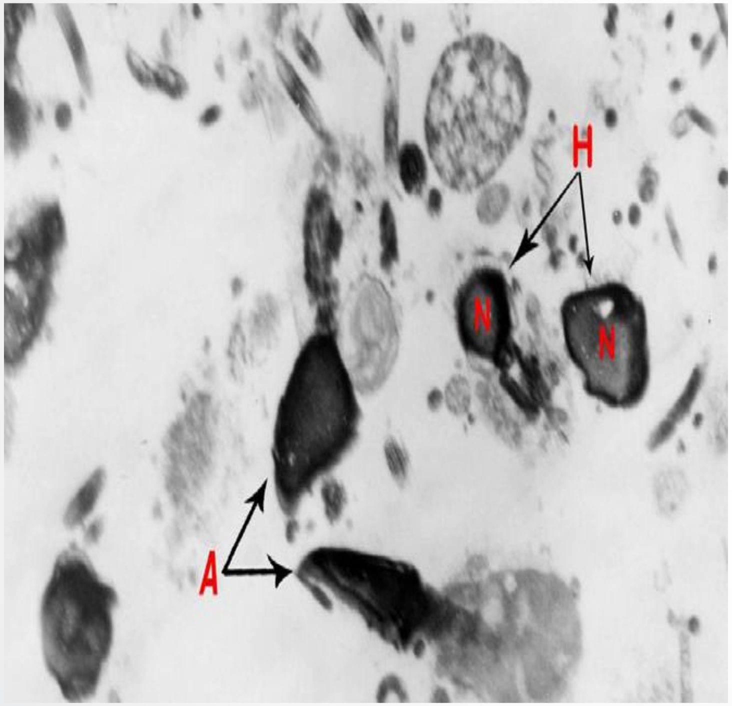
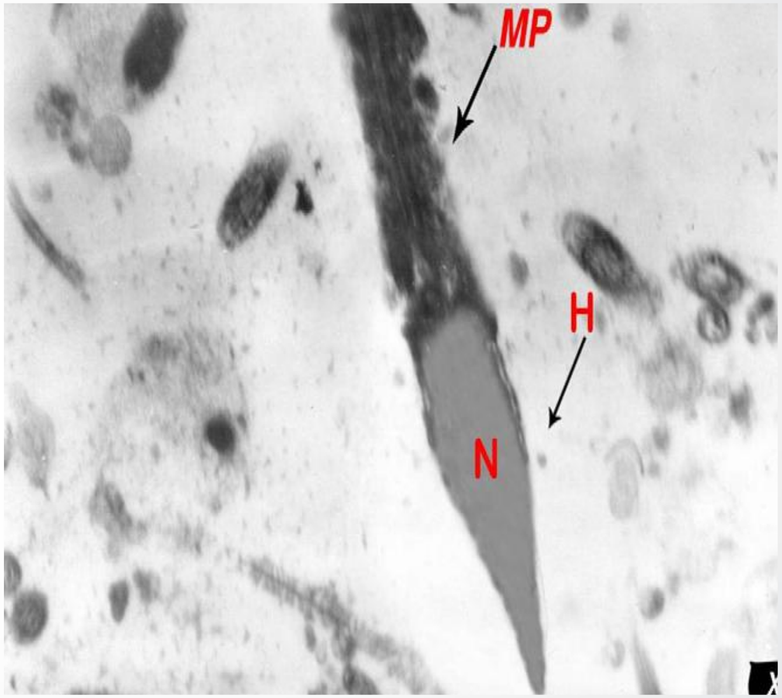


Figure2. Electron photomicrograph of longitudinal section of normal human sperm of healthy group displaying, normal organization of the nucleus (N) and acrosome (A). The neck region are showing two opposite planes parallel and perpendicular to the long axis of the proximal centriole (C). Note, the segmented mitochondria (M) with well defined mitochondrial sheath (MS) and helicoidally distribution of mid piece mitochondria

Teratozoospermic group

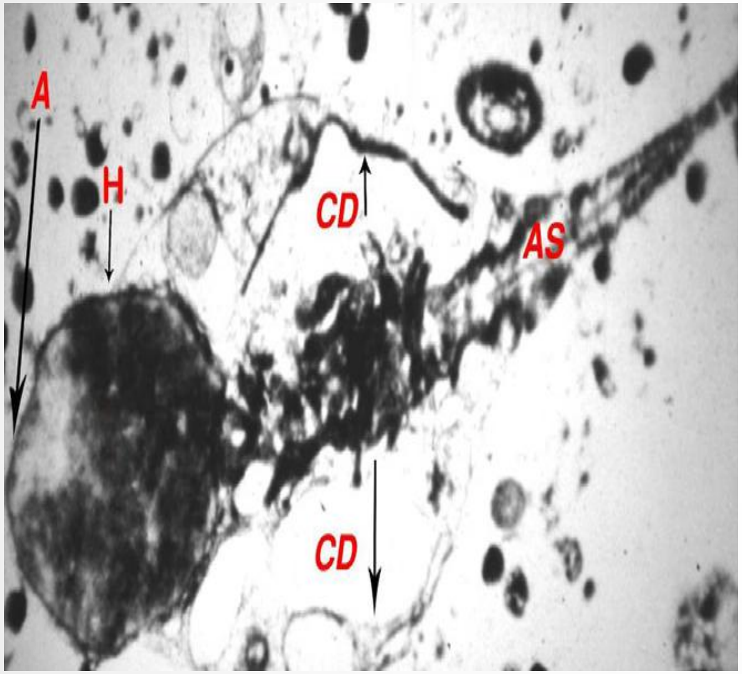


Electron photomicrograph of human sperm of terato group before treatment showing hypoplasia; small head (H), abnormal acrosome (A) and sphericity of the nucleus (N). ·4600.

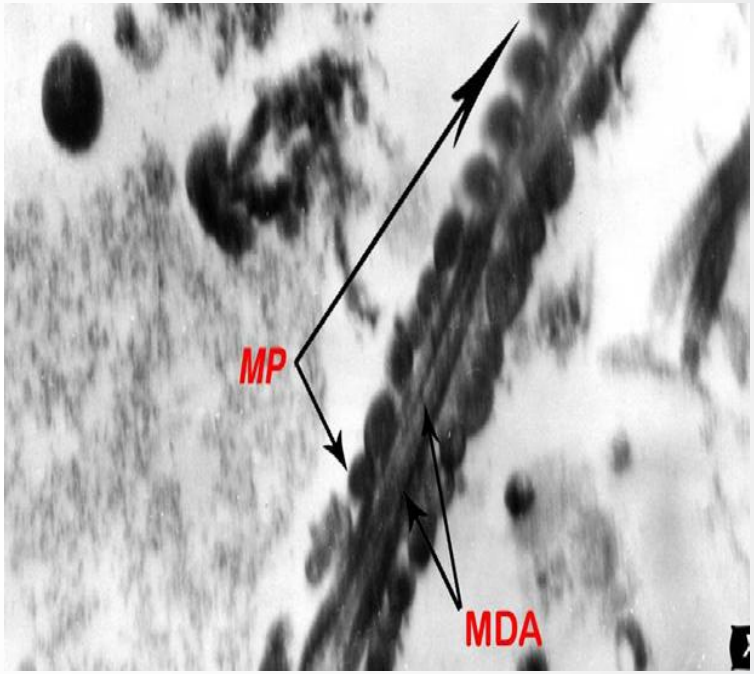


Electron photo micrograph of longitudinal section of human sperm of terato group after treatment demonstrating, normal head (H), normal nuclear pattern (N) and normal arrangement of the mitochondrial structure of the mid piece (MP). ·5000.

Athenozoospermic group

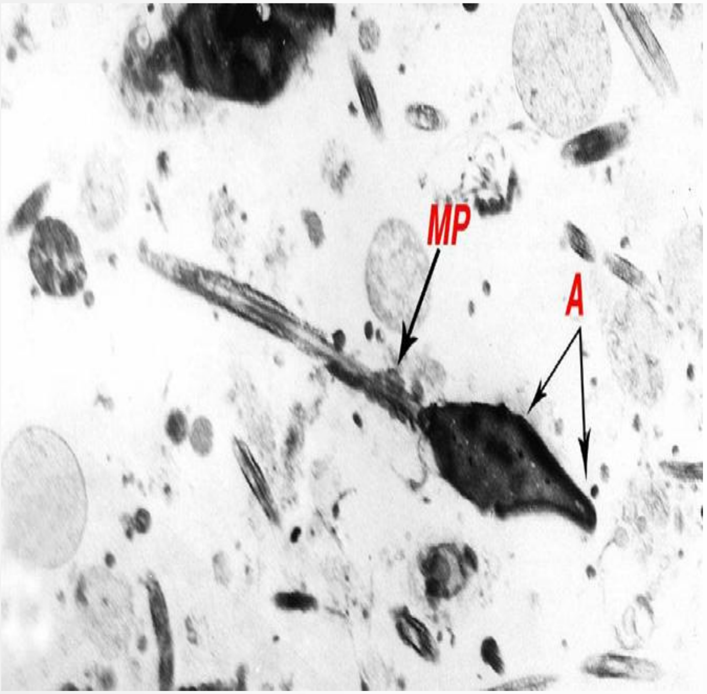


Electron photomicrograph of longitudinal section of human sperm of atheno group before treatment showing, abnormal rounded head (H), degenerated acrosome (A), big cytoplasmic droplet (CD) and abnormal axonemal structure (AS). ·3000.

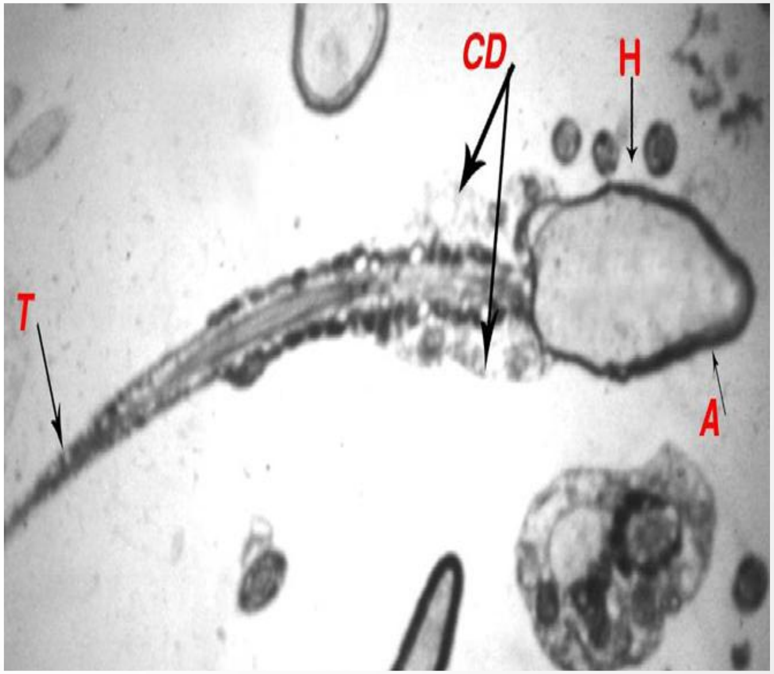


Electron photomicrograph of longitudinal section of human sperm of atheno group after treatment displaying, normal mid piece (MP) and regular microtubule doublets of the axoneme (MDA). ·13,000.

Oligozoospermic group



Electron photomicrograph of longitudinal section of human sperm of oligo group before treatment displaying, malformed acrosome (A) and thin mid piece (MP) with defected mitochondria (M). ·4600.



Electron photomicrograph of longitudinal section of human sperm of oligo group after treatment demonstrating, normal head (H), normal acrosome cap (A) and well defined tail (T). But the most proximal segment of the flagellum is enveloped by cytoplasmic droplet (CD). ·6000.

L-methionine

- ✓ Methionine is one of 8 essential amino acid, it is used to make peptides for sperm’s membrane & overall structure
- ✓ Methionine has antioxidant effect that fights harmful free radicals, it helps body to eliminate fats
- ✓ Methionine is used for the treatment of premature ejaculation & also used for chronic depression
- ✓ Methionine Increases sperm count and plays a role in sperm motility

Lycopene

- Protects against lipid and DNA oxidation, due to its antioxidant effect & neutralizes reactive oxygen species that can cause male infertility
- Lycopene is found in high levels in seminal fluid
- Men who have poor quality of sperms can benefit from Lycopene, because it limits oxidative stress & it is used for treatment of idiopathic male infertility
- Lycopene has been shown in one of the "Harvard University" studies to reduce risk of prostate cancer & heart diseases

Lycopene improves the distorted ratio between AA/DHA in the seminal plasma of infertile males and increases the likelihood of successful pregnancy

- The subjects were subdivided into 21 normozoospermic males from couples with idiopathic infertility and 23 males with semen abnormalities.

- Prior to the treatment with lycopene, the AA/DHA ratios in both subgroups of patients were significantly higher than in fertile controls and improved following treatment with lycopene.

Radka Filipcikov, et al. 2015 Mar; 159(1):77 82.77

Lycopene improves the distorted ratio between AA/DHA in the seminal plasma of infertile males and increases the likelihood of successful pregnancy

- Seven spontaneous pregnancies (16%) occurred before the scheduled IVF treatment

- 15 couples (42%) achieved pregnancy after IVF

Conclusions. Three months of treatment with lycopene led to a significant improvement in the AA/DHA ratio in seminal plasma of males from infertile couples and facilitated the spontaneous as well as IVF conception.

Radka Filipcikov, et al. 2015 Mar; 159(1):77-82.77

Efficacy of coenzyme Q10 on semen parameters, sperm function and reproductive hormones in infertile men

212 infertile men with IOAT

Group 1
300mg CoQ10/ day

Group 2
Placebo regimen

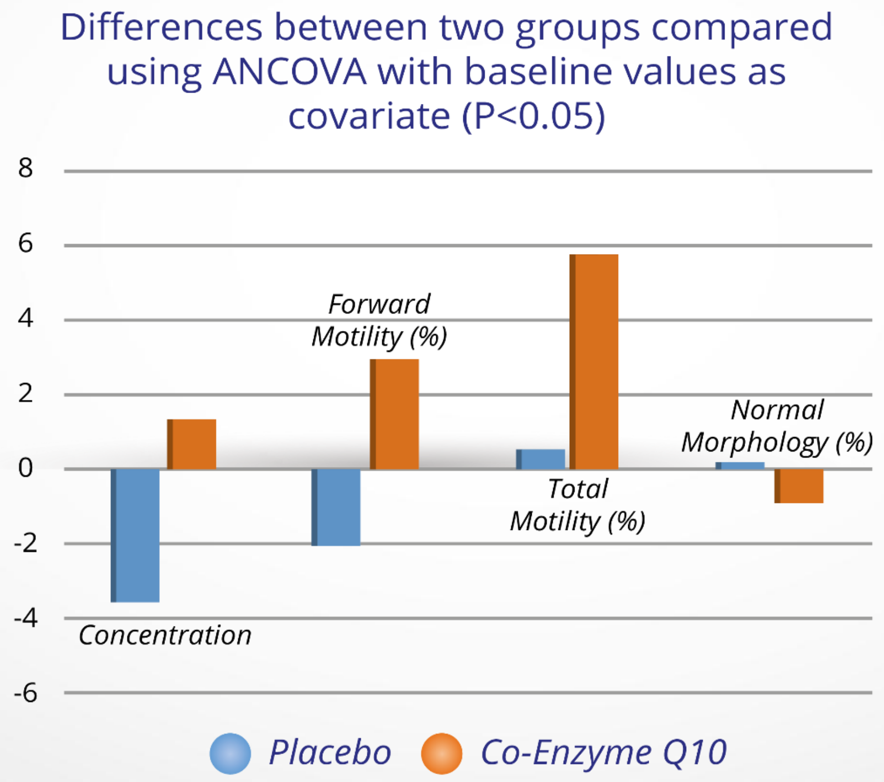
Safarinejad, M. R. (2009). *Journal of Urology, the, 182*(1), 237-248.

RESULTS CoQ10 group

- ✓ Significant improvement in sperm density and motility was evident
- ✓ Using the Kruger classification sperm morphology evaluation revealed an increase in the percent of normal forms
- ✓ A positive correlation was found between treatment duration with coenzyme Q10 and sperm count, as well as with sperm motility and sperm morphology
- ✓ Significant decrease in serum FSH and LH at the 26-week treatment phase
- ✓ By the end of the treatment phase the mean +/- SD acrosome reaction had increased from 14% +/- 8% to 31% +/- 11%

Coenzyme Q10 supplementation resulted in a statistically significant improvement in certain semen parameters

Three-month supplementation with CoQ10 in OAT infertile men can attenuate oxidative stress in seminal plasma and improve semen parameters and antioxidant enzymes activity



Nadjarzadeh, A. et al, (2014). Andrologia, 46(2), 177-183.

Ubiquinol was significantly effective in men with unexplained oligoasthenoteratozoospermia for improving sperm density, sperm motility and sperm morphology

228 infertile men with IOAT

Group 1
200mg Ubiquinol daily

Group 2
Placebo regimen

A double-blind, placebo controlled, randomized study.

J Urol 2012 Aug;188(2):526-31. doi: 10.1016/j.juro.2012.03.131. Epub 2012 Jun 15.

Anti-Oxidant	Sperm Count	Sperm Motility	Sperm Morphology	DNA Damage
CoQ10	√	√	√	
Vit E	√	√	√	√
Sel		√	√	√
Vit C				√
Zinc				√
A-L-C		√	√	

Multivitamin and micronutrient treatment improves semen parameters of azoospermic patients with maturation arrest.

Singh AK, et al 2010.

- The study was undertaken to evaluate the efficacy of multivitamin and micronutrient supplementation in azoospermic patients with maturation arrest.

- **A total of 35 azoospermic patients showing maturation arrest on testicular biopsy were recruited in this study.**

- The patients were divided into two groups. Untreated group (n=11) without any treatment and treated group (n=24) who received multivitamins, micronutrients and co-enzyme Q10.

Multivitamin and micronutrient treatment improves semen parameters of azoospermic patients with maturation arrest.

Singh AK, et al 2010.

- The results showed **reduction in liquefaction time and relative viscosity of the semen** in the treated group. Further, in treated group there was appearance of spermatozoa (4.0 million/ml) exhibiting progressive motility (7%) and normal morphology (6%), even in the first follow up visit.

- **The sperm count, motility and normal morphology increased significantly on subsequent visits. Within 3 months (3 visits) 2 pregnancies were reported.**

- These observations indicate that multivitamin and micronutrient supplementation improve the qualitative and quantitative parameters of seminogram in patients with azoospermia of maturation arrest.

- **Daily Dose:**

4 capsules in divided doses preferably after meals

- **Duration of use:**

  3 months

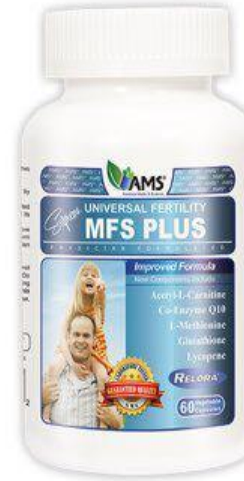
- **Semen Analysis** at baseline & after 3 months

RESEARCH

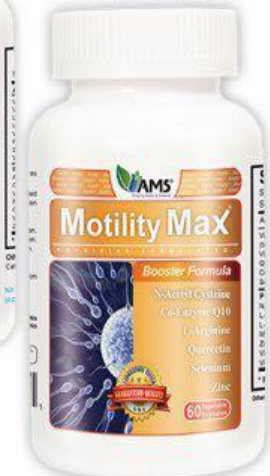
The Semen pH Affects Sperm Motility and Capacitation:

Published: July 14, 2015
Center for Reproductive Medicine, Jinling Hospital
Clinical School of Medical College, Nanjing University Nanjing, People's Republic of China

AMS
America Medic & Science



Lycopene



Macuna pruriens

METHOD

A total of 136 male volunteers participated in this study, with a mean age of 27.54 ± 3.98
The sperm subjected to analysis were from healthy male volunteers.The semen samples were obtained by masturbation after at least 3 days of abstinence. Purified sperms were resuspended in sperm nutrition solution of pH 5.2, 6.2, 7.2 and 8.2, and incubated for 15, 30, 60, 90 and 120 min respectively.

RESULTS

1- Sperm total **motility**, progressive motility were similar in pH 7.2 and 8.2, but **decreased in pH 5.2 and 6.2 solutions.**
2- The viability of sperm was decreased in acidic environments
3- Acidic environments reduced [Ca2+]i during human sperm capacitation.
4- Sperm penetration decreased in acidic environments.
5- Sperm Na+/K+-ATPase activity decreased in acidic environments.

CONCUSION

In conclusion, we found slightly alkaline conditions seem to stimulate human sperm motility and capacitation. The dramatically decreased Na+/K+-ATPase activity caused by acidic envi- ronment could be the reason of lower sperm motility and Ca2+ level, which ultimately affected pregnancy. However, the mechanisms of pH affecting Na+/K+-ATPase activity in human sper- matozoa are not completely elucidated. In addition to affecting Na+/K+-ATPase, acidic envi- ronment may be able to damage the sperm cell membrane directly, or to increase the active oxygen content, thus affecting sperm motility and capacitation, which warrant further explorations.

THANK YOU